Cryptography Notes

Oli

Feb - Jun, 2026

Contents

1. Algebra

1.1. Notation

- $\boldsymbol{Z}$ = 整數 integers
- $\boldsymbol{Z_n} = \{0, 1, 2, ..., n-1\}$ ($0 \sim n-1$的整數)
- $\boldsymbol{Z_n^*} = \{a \in Z_n \mid \gcd(a, n) = 1\}$ (Z_n 裡面和 n 互質的)

> eg. $Z_{12}^* = \{1, 5, 7, 11\}$

- $\boldsymbol{N}$ = non-negative integers (有 0)
- $\boldsymbol{P}$ = positive integers (無 0)
- $\boldsymbol{Q}$ = rational numbers
- $\boldsymbol{R}$ = real numbers
- $\boldsymbol{C}$ = complex numbers
- $\boldsymbol{Q^*}, \boldsymbol{R^*}, \boldsymbol{C^*}$ = **non-zero** rational, real, complex numbers
- $\boldsymbol{GL_n(R)}$
  - $n \times n$ 的可逆 (invertible) 方陣
  - 裡面的元素 (entries) 都在 R 裡
- $\boldsymbol{SL_n(R)}$
  - $n \times n$ det 為 1 的方陣 (det 為 1 所以也可逆)
  - 裡面的元素都在 R 裡、是 $GL_n(R)$ 的 subset

1.2. Group

1.2.1. Definition

group $(G, *)$ 是一個集合 G 跟一個運算 $*$ 所構成的 (中文是群)
一個 group 要滿足四個條件：

1. **Closure 封閉性**
   $a * b \in G \ \forall a, b \in G$
   $a * b$ 後的答案還是得在 G 裡面
2. **Associativity 結合律**
   $a * (b * c) = (a * b) * c \ \forall a, b, c \in G$
3. **Identity**
   $\exists e \in G \ s.t. \ a = a * e = e * a$
   找得到一個 identity e 讓 a 和 e 運算或 e 和 a 運算後還是自己

   > $*$ 通常是加或乘
   > 加法的 identity e 通常是 0 (0 加自己或自己加 0 都還是自己)
   > 減法的 identity e 通常是 1 (1 乘自己或自己乘 1 都還是自己)

4. **Inverse 反元素**
   $\forall a \in G, \exists b \in G \ s.t. \ a * b = b * a = 1$
   找得到跟和 a 運算完會變 identity 的 inverse b

Group 的 Example

- Z, Q, R, C with $+$
- Q^*, R^*, C^* with $\times$
- $5Z = \{5a \mid a \in Z\}$ with $+$
- $\{1, -1\}$ with $\times$
- $Z_6 = \{0, 1, 2, 3, 4, 5\}$ with $+ \bmod 6$
- Z_7^* with $\otimes$ ($\otimes$ 表示 $\times \bmod p$, p 是 Z 的下標 所以在這邊 $\otimes = \times \bmod 7$)
- $\{(x, y) \in R^2 \mid y^2 = x^3 + ax + b\} \cup \{\infty\}$ with point addition (加法運算：兩個點相加) and point doubling (兩倍運算：一個點的兩倍) on elliptic curves

不符合 Group 的 Example

- Z_7 with $+ \Rightarrow$ not closed
- P with $+ \Rightarrow$ no identity
- Z with $- \Rightarrow$ no nssociativity
- $2Z + 1 = \{2a + 1 \mid a \in Z\}$ (奇數) with $\times \Rightarrow$ no inverse

1.2.2. Proposition

1. The identity e of G is unique

> Proof
> 假設有兩個 identity e_1 跟 $e_2 \Rightarrow e_1 = e_1 * e_2 = e_2$

2. The inverse b of each $a \in G$ are unique, denoted as a^{-1}

> Proof
> 假設 b 跟 c 都是 a 的 inverse $\Rightarrow b = b * e = b * (a * c) = (b * a) * c = e * c = c$

3. $\left(a^{-1}\right)^{-1} = a \ \forall a \in G$
4. $(a * b)^{-1} = b^{-1} * a^{-1} \ \forall a, b \in G$
5. $\forall a, b, c \in G$
   a. $a * x = b$ and $y * a = b$ have unique solution

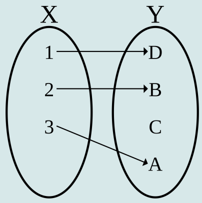
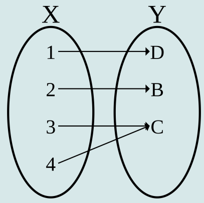
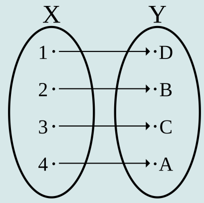
   > Proof
   > $a^{-1} * (a * x) = a^{-1} * b \to \left(a^{-1} * a\right) * x = a^{-1} * b \to e * x = a^{-1} * b$
   > $\Rightarrow x = a^{-1} * b$

   b. $a * b = a * c$ 代表 $b = c$ and $a * b = c * b$ 代表 $a = c$

   > Proof
   > 兩邊同乘 a^{-1} 然後套一下四個性質

1.2.3. Permutation

A **permutation** is a bijective function $f : T \to T$ on a set T into itself

injective 單射 (one-to-one)	**surjective 滿射** (onto)	**bijective 雙射** (one-to-one and onto)
		

Permutation 的 Example

$T = \{1, 2, 3\}$ with $f(1) = 2$, $f(2) = 3$, $f(3) = 1$ is a permutation represented by the array

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

就是矩陣裡面上面那排的經過 f 函數會被換成下面那排的

Remark

Every block cipher with a specific key is a permutation

- $DES_k : \{0,1\}^{64} \to \{0,1\}^{64}$ (block size = 64 bits)
- $AES_k : \{0,1\}^{128} \to \{0,1\}^{128}$ (block size = 128 bits)

1.2.4. Symmetric Group

A **symmetric group** S_n is the set of all permutations on $\{1, 2, 3, ..., n\}$ with $\circ$ (compositon)

$(f \circ g)(x) = f(g(x))$ (合成函數 先運算後面的再運算前面的)

Symmetric Group 的 Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}$$

$$S_3 = \left\{ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} \right\}$$

可以把上面那些東西簡寫：$S_3 = \{e, (12), (13), (23), (123), 132\}$

- $(123)^{-1} = (321) = (132)$

Remark

The S-box of AES is a permutation in S_{256}

1.2.5. Abelian Group

A group $(G, *)$ is **commutative** or **abelian** if $a * b = b * a \ \forall a, b \in G$

Abelian Group 的 Example

- Z, Q, R, C with $+$
- $Z_9^* = \{1, 2, 4, 5, 7, 8\}$ with $\otimes$
  在 Z_9^* 中 2 和 5 的次方都會跑遍所有元素 所以他們是 Z_9^* 的生成元 (generator)
  以 2 為例 : $1 = 2^6, 2 = 2^1, 4 = 2^2, 5 = 2^5, 7 = 2^4, 8 = 2^3$
- Z_p^* with $\otimes$ for every prime p

Non-Abelian Group 的 Example

只要找到任意一個 $a * b \neq b * a$ 就是了

- for $n \geq 3$, the symmetric group S_n is not abelian :

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \circ \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

- with the operation of matrix multiplication, each of th following is not abelian :
  - A general linear group $GL_n(R)$
  - A special linear group $SL_n(R)$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

  the above is in $SL_2(Q) \subset GL_2(Q)$

1.2.6. Cyclic Group

A group $(G, *)$ is **cyclic** if there exists a generator $g \in G$ $s.t.$ every $a \in G$ is of the form $a = g * g * g * \dots * g$ (n copies) for some $n \in Z$ (n 可以 < 0)

- 2 cpoies $\Rightarrow a = g * g$
- -2 copies $\Rightarrow a = g^{-1} * g^{-1}$

Cyclic Group 的 Example

- $(Z, +)$ with generators 1 and -1
- $(Z_7^*, \otimes)$ with generator 3
- $(Z_9^*, \otimes)$ with generators 2 and 5

不是 Cyclic Group 的 Example

- $(Q, +)$
- $(Z_8^*, \otimes)$ (這種有 4 個元素而且都不是 generator 的 group 叫 Klien 4)

1.2.7. Group Order

The **order** of a group $(G, *)$ (denoted as $|G|$ or $\#G$) is the number of the elements in G, or the **cardinality** of G

Group Order 的 Example

- $|S_3| = 6, |S_4| = 24, |S_n| = n!$
- $|Z_p| = p, |Z_p^*| = p - 1$ for any prime p
- $|Z_9^*| = 6$
- $|Z_n^*| = |\{a \in Z_n \mid \gcd(a, n) = 1\}| = \phi(n)$
  $\phi(n)$ 叫 euler phi-function
- 用來 construct AES 的 S-box 的 $|GL_8(Z_2)|$ =??

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \in GL_8(Z_2)(= SL_8(Z_2))$$

A group $(G, *)$ is **finite** if $|G| < \infty$, and is **infinite** if $|G| = \infty$

Finite / Infinite 的 Example

- $|Z| = \infty, |Q| = \infty, |R| = \infty, |C| = \infty$
  - $|Z|$ and $|Q|$ have the same cardinality, which is countable
  - $|R|$ and $|C|$ have the same cardinality, which is uncountable

1.2.8. Subgroup and Coset

A subset H of a group $(G, *)$ is a **subgroup** of G if H is itself a group under $*$ in G
A subgroup H of a group $(G, *)$ is a subgroup of G iff

1. $a * b \in H \ \forall a, b, \in H$
2. $a^{-1} \in H \ \forall a \in H$

Subgroup 的 Example

- $(5Z, +)$ is a subgroup of $(Z, +)$
- $(Q^*, \times)$ is a subgroup of $(R^*, \times)$

If H is a subgroup of a group G, and $g \in G$, then

- $gH := \{g * h \mid h \in H\}$ is a **left coset of H** in G
- $Hg := \{h * g \mid h \in H\}$ is a **right coset of H** in G
- for abelian groups, $gH = Hg$

Remark

In general, Left coset $\neq$ Right coset

Coset 的 Example

- $(123)\{e,(12)\} = \{(123),(13)\}$ ((123) 是 S_3 的元素、$\{e,(12)\}$ 是 S_3 的 subgroup)
  $\{e,(12)\}(123) = \{(123),(23)\}$
  $\Rightarrow L \neq R$
- $(12)\{e,(123),(132)\} = \{(12),(23),(13)\}$
  $\{e,(123),(132)\}(12) = \{(12),(13),(23)\}$
  $\Rightarrow L = R$

$g_1 H = g_2 H \Leftrightarrow g_1^{-1} g_2 \in H$

這個性質的 Example

$(123)\{e,(12)\} = (13)\{e,(12)\} \Leftrightarrow (123)^{-1}(13) = (132)(13) = (12) \in \{e,(12)\}$

1.2.9. Lagrange Theorem

If H is a subgroup of a finite group G, then $|H|$ divides $|G|$

H is a subgroup of a finite group G, then $[G:H] := \frac{|G|}{|H|}$ is the **index** of H in G

For $g \in G$, the **order** of g, denoted by $o(g)$, is the smallest positive integer n $s.t.$ $g^n = e$

$\rightarrow o(g)$ divides $|G|$: $<g> = \left\{g, g^2, g^3, ..., g^{o(g)-1}, g^{o(g)}(=e)\right\}$ is a subgroup of G

這個性質的 Example

- $\{e,(123),(132)\}$ is a subgroup of S_3, $3 \mid 6$
- $\{1,8\}$ is a subgroup of $(Z_9^*, \otimes)$, $2 \mid 6$
- $SL_2(Z_7)$ is a subgroup of $GL_2(Z_7)$
  $|GL_2(Z_7)| = (7^2-1)(7^2-7)$, $|SL_2(Z_7)| = \frac{|GL_2(Z_7)|}{6}$ (下面有解釋)
- $T_2(Z_7) = \{M \in GL_2(Z_7) \mid M \text{ is upper triangular }\}$ is a subgroup of $GL_2(Z_7)$
  $|T_2(Z_7)| = 6 \times 6 \times 7$

$|SL_2(Z_7)| = \frac{|GL_2(Z_7)|}{6}$ 的解釋 (可以推廣至 $|SL_n(Z_p)|$、$|GL_n(Z_p)|$)

1. Let $H = SL_2(Z_7) = \left\{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \middle| \ a_{ij} \in Z_7, a_{11}a_{22} - a_{12}a_{21} = 1\right\}$
2. For $t = 2,3,4,5,6$, define the left coset $H_t = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} H$, then
   a. $H_t = \{M \in GL_2(Z_7) | \det(M) = t\}$
   b. H and H_t's are mutually disjoint (兩兩互斥：任兩個集合中沒有共同元素)
   c. $f_t : H \rightarrow H_t$ defined by $f_{t(M)} = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} M$ is bijective
   d. $|H| = |H_t|$
   e. $GL_2(Z_7) = H \cup H_2 \cup H_3 \cup H_4 \cup H_5 \cup H_6$
3. $|GL_2(Z_7)| = |H| + |H_2| + |H_3| + |H_4| + |H_5| + |H_6| = 6 \times |SL_2(Z_7)|$

1.2.10. Homomorphism and Isomorphism

$(G, *)$ and $(H, \star)$ are two groups

- A function $f : G \to H$ is a **homomorphism** (同態) if
  - $f(a * b) = f(a) \star f(b)\ \forall a, b \in G$
- A function $f : G \to H$ is a **isomorphism** (同構) (G is isomorphic to H / $G \cong H$) if
  - $f(a * b) = f(a) \star f(b)\ \forall a, b \in G$
  - f is bijective

Homomorphism 的 Example

- $f : (Z, +) \to (Z, +)$ is a homomorphism with $f(1) = 2$
  - injective (one-to-one), not surjective (onto)
- $f : (Z, +) \to (Z_7^*, \otimes)$ is a homomorphism with $f(1) = 3$
  - surjective (onto), not injective (one-to-one)

Isomorphism 的 Example

- $f : (C^*, \times) \to (C^*, \times)$ defined by $f(a + bi) = a - bi$
- $f : (Z_6, + \bmod 6) \to (Z_7^*, \times \bmod 7)$ is a isomorphism defined with $f(1) = 3$
  - $f(2) = f(1+1) = 2, f(3) = f(1+1+1) = 6, f(4) = 4, f(5) = 5, f(0) = 1$
  - $f(0) = 1$ 因為 $f(0) = f(0+0) = f(0) \times f(0)$ 兩邊同乘 $f(0)^{-1}$ 得到 $f(0) = 1$
  - 思考：如果 $f(1) = 5, f(1) = 2$ 會怎麼樣

1.3. Ring

1.4. Field

2. Arithmatic

2.1. Modular Function

Floor and Ceiling

- the **floor** $\lfloor x \rfloor$ of $x \in R$ is the largest integer $\leq x$
- the **ceiling** $\lceil x \rceil$ is the smallest integer $\geq x$

eg. $\lfloor e \rfloor = 2, \lceil e \rceil = 3, \lfloor -3.14 \rfloor = -4, \lceil -3.14 \rceil = 3$
eg. $25 = 3_{\text{(quotion)}} \times 7_{\text{(divisor)}} + 4_{\text{(remainder)}} \Rightarrow 25 \bmod 7 = 25 - \lfloor \frac{25}{7} \rfloor \times 7$

Modular Function

$m, n \in Z, m > 0$, define $\boldsymbol{n} \,\mathbf{mod}\, \boldsymbol{m} = n - \lfloor \frac{n}{m} \rfloor \times m$ (也就是 $\frac{n}{m}$ 之後的餘數)

eg. 58 in the base 3 representation :
$\lfloor \frac{58}{3} \rfloor = 19, 58 \bmod 3 = 1$
$\lfloor \frac{19}{3} \rfloor = 6, 19 \bmod 3 = 1$
$\lfloor \frac{6}{3} \rfloor = 2, 6 \bmod 3 = 0$
$\lfloor \frac{2}{3} \rfloor = 0, 2 \bmod 3 = 2$
$\Rightarrow (58)_{10} = (2011)_3$

Cayley Tables

Define $\oplus$ and $\otimes$ on $Z_6 = \{0, 1, 2, 3, 4, 5\}$ as :

$a \oplus b = (a + b) \textsc{ mod } 6$

$\oplus$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$a \otimes b = (a \times b) \textsc{ mod } 6$

$\otimes$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

$\Rightarrow Z_6$ 不是 field 因為 2 沒有乘法反元素

2.2. GCD and Euclidean Algorithm

2.2.1. Greatest Common divisor (GCD)

Assume $a, b \in Z, a \neq 0$ or $b \neq 0$:

- $d \neq 0$ is a **common divisor** of a and b if $d \mid a$ and $d \mid b$
- The **greatest common divisor** $d = \gcd(a, b)$ is the largest of the common divisors

> eg. divisors of 20 : $\pm 1, \pm 2, \pm 5, \pm 10, \pm 20$

2.2.2. Proposition

if $a \in P$

- $\gcd(a, a) = a$
- $\gcd(a, 0) = a$

if $a, b \in Z, a \neq 0$ or $b \neq 0$

- $\gcd(a, b) = \gcd(|a|, |b|)$
- $\gcd(a, b) = \gcd(b, a)$

$a, b \in Z, a \neq 0$ or $b \neq 0$, then $\gcd(a, b) = \gcd(a + kb, b)$ for any $k \in Z$ (這是 Thm 不是 Prop)

> Proof
> Define $A = a$ 和 b 的公因數的 set, $B = a + kb$ 和 b 的公因數的 set
> (i) $A \subset B$: 假設 $d \in A \Rightarrow d \mid a, d \mid b \Rightarrow a = xd, b = yd$ for some $x, y \in Z$
> $\Rightarrow a + kb = xd + k(yd) = (x + ky)d \Rightarrow d \mid (a + kb) \Rightarrow d \in B$
> (ii) $B \subset A$:$\Rightarrow c \in B$，這樣 $c \mid (a + kb), c \mid b \Rightarrow a + kb = xc, b = yc$ for some $x, y \in Z$
> $\Rightarrow a = xc - kb = xc - k(yc) = (x - ky)c \Rightarrow c \mid a \Rightarrow c \in A$
> $\Rightarrow A = B, \gcd(a, b) = \gcd(a + kb, b)$

Corollary : 如果 $b > 0$, then $\gcd(a, b) = \gcd(b, a \bmod b)$

> Proof
> $\gcd(a, b) = \gcd\left(a - \lfloor \frac{a}{b} \rfloor \times b, b\right)_{\text{Thm}} = \gcd(a \bmod b, b)_{\text{Def}} = \gcd(b, a \bmod b)_{\text{Prop}}$

2.2.3. Euclidean Algorithm

歐幾里德演算法 aka 輾轉相除法

直接算

$\gcd(325, 234)$
$= \gcd(234, 325 \bmod 234)$
$= \gcd(234, 91)$
$= \gcd(91, 234 \bmod 91)$
$= \gcd(91, 52)$
$= \gcd(52, 91 \bmod 52)$
$= \gcd(52, 39) = \ldots$

直式

1	325	234	2
	234	182	
1	91	52	1
	52	39	
3	39	13	
	39		
	0		

Pseudo Code

normal way

```
if b = 0
    gcd = |a|
else
    while b != 0
        c = b
        b = a mod b
        a = c
return gcd
```

recursive way

```
if b = 0
    gcd = |a|
else
    gcd = gcd(b, a % b)
return gcd
```

2.3. Extended GCD Algorithm

Algorithm : Given $a, b \in Z$, find $x, y \in Z$ s.t. $ax + by = \gcd(a, b)$
Application : b是質數$\Rightarrow \gcd(a, b) = 1 \Rightarrow a, b$ 互質
法一 $\Rightarrow a \bmod b$ 乘法反元素存在 (假設叫 x) $\Rightarrow ax \equiv 1 (\bmod b)$
法二 $\Rightarrow ax + by = 1 \Rightarrow ax + by (\bmod b) = 1 (\bmod b) \Rightarrow ax \equiv 1 (\bmod b)$

eg. $a = 100, b = 35$
$100 = 2 \times 35 + 30$
$35 = 1 \times 30 + 5$
$30 = 6 \times 5 + 0$
$\Rightarrow \gcd(100, 35) = 5 = 35 - 30 = 35 - (100 - 2 \times 35) = (-1) \times 100 + (3) \times 35$
$\Rightarrow x = -1, y = 3$
先用輾轉相除法找 gcd 再原路代回去

Theorem : given $a, b \in Z$, there $\exists\ x, y \in Z$ for the **linear combination** $ax + by = \gcd(a, b)$

Proof
take $r_0 = a = a \times 1_{[=x_0]} + b \times 0_{[=y_0]} = ax_0 + by_0$
$r_1 = b = a \times 0_{[=x_1]} + b \times 1_{[=y_1]} = ax_1 + by_1$
for $i > 0$, let $q_{i+1} = \lfloor \frac{r_{i-1}}{r_i} \rfloor$
$r_{i+1} = r_{i-1} - r_i q_{i+1}$
$x_{i+1} = x_{i-1} - x_i q_{i+1}$
$y_{i+1} = y_{i-1} - y_i q_{i+1}$
assume $r_{i-1} = ax_{i-1} + by_{i-1}$ and $r_i = ax_i + by_i$
then $r_{i+1} = r_{i-1} - r_i q_{i+1} = (ax_{i-1} + by_{i-1}) - (ax_i + by_i) q_{i+1} = ax_{i+1} + by_{i+1}$
$\Rightarrow$ 由數學歸納法得證

Algorithm (matrix version)

1 set $\begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix} = \begin{bmatrix} a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}$
2 while $r \neq 0$
3 set $\begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{r_0}{r_1} \rfloor \end{bmatrix} \begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix}$
4 return $[d, x, y] = [r_0, x_0, y_0]$

2.4. Stein's GCD Algorithm / Stein's Recursion

three least-significant-bit GCD properties

- a, b both even $\Rightarrow \gcd(a, b) = 2\gcd(\frac{a}{2}, \frac{b}{2})$
- a is even, b is odd $\Rightarrow \gcd(a, b) = \gcd(\frac{a}{2}, b)$
- a, b both odd $\Rightarrow \gcd(a, b) = \gcd(\frac{a-b}{2}, b)$

一直遞迴下去直到找到 gcd 為止
因為除以 2 只是往右 shift 一格而已，沒有一般的除法麻煩 $\Rightarrow$ 目前最快的 gcd algorithm

2.5. Many Theorems of Arithmatic

- **Theorem :** $\gcd(a, b)$ is the smallest positive integer that can be written as $d = ax + by$ where $x, y \in Z$
- **Definition :** a and b are **relatively prime** ($a \perp b$) if $\gcd(a, b) = 1$
- **Theorem :** If $a \mid (bc)$ and $a \perp b$ then $a \mid c$

> Proof
> $a \perp b \Rightarrow ax + by = 1$ for some $x, y \in Z$
> multiply by $c \Rightarrow acx + bcy = c \Rightarrow a \mid (bc) \Rightarrow a \mid (bcy) \Rightarrow a \mid (acx + bcy) = c$

- **Corollary :** p is a prime and $p \mid (ab)$, then $p \mid a$ or $p \mid b$

> Proof
> 兩種可能 $\Rightarrow$ 1. $p \mid a \Rightarrow$ done, 2. $p \nmid a \Rightarrow \gcd(p, a) = 1 \Rightarrow p \mid b$ by theorem

- **Corollary :** p is a prime and $p \mid (a_1 a_2 a_3 \ldots a_k)$, then $p \mid a_i$ for some i

> Proof
> 一直做上一個 corollary 的動作就好

- **Theorem (The Fundamental Theorem of Arithmatic) :** Every integer $a > 1$ is itself prime or can be written unique as a product of primes

> Proof
> Existence : 一直做上一個 corollary 的動作就好
> Uniqueness : 用上一個 corollary

- **Definition :** Unique Factorization Domain (UFD) 代表有唯一質因數分解的集合

> UFD 的 Example
> - Z is a UFD
> - $Z\left[\sqrt{-5}\right] = \left\{a + b\sqrt{-5} \mid a, b \in Z\right\}$ is not UFD, 例如 6 就有兩種分解法
> $\Rightarrow 6 = 2 \times 3 = \left(1 - \sqrt{-5}\right)\left(1 + \sqrt{-5}\right) \Rightarrow$ 都已經沒辦法分解了 (irreducible)

- **Definition :** The least common multiple $\operatorname{lcm}(a, b)$ of $a, b \in P$ is the smallest $m \in P$ s.t. $a \mid m$ and $b \mid m$
- **Theorem :** $a, b \in P$, then $a \times b = \gcd(a, b) = \operatorname{lcm}(a, b)$

2.6. Modular Arithmatic

2.6.1. Congruence Relation (同餘)

Given $a, b \in Z, m \in P$, then a is congruent to b modulo m ($a \equiv b (\operatorname{mod} m)$)

- $a \equiv b$ iff $m \mid (a - b)$

2.6.2. Proposition

- if $a, b, c \in Z, m, d \in P$, then
  - $a \equiv a (\operatorname{mod} m)$
- if $a \equiv b (\operatorname{mod} m)$, then
  - $b \equiv a (\operatorname{mod} m)$
  - $a + c \equiv b + c (\operatorname{mod} m)$
  - $ac \equiv bc (\operatorname{mod} m)$
  - $a^d \equiv b^q (\operatorname{mod} m)$

> Proof
> $a \equiv b (\operatorname{mod} m) \Leftrightarrow m \mid (a - b)$
> $\Rightarrow m \mid (a - b)\left(a^{d-1} + a^{d-2}b + \ldots + b^{d-1}\right) = a^d - b^d$

- if $a \equiv b$ and $b \equiv c (\operatorname{mod} m)$, then $a \equiv c (\operatorname{mod} m)$
- if $d \equiv \gcd(a, m)$, $ab \equiv ac (\operatorname{mod} m)$, then $b \equiv c \left(\operatorname{mod} \frac{m}{d}\right)$

> Proof
> we have $d = ax + my$ for some $x, y \in Z$
> Hence $\left(\frac{a}{d}\right)x + \left(\frac{m}{d}\right)y = 1$ where $\frac{a}{d}, \frac{m}{d} \in Z$
> multiply by $(b - c) \Rightarrow \left(\frac{a}{d}\right)(b - c)x + \left(\frac{m}{d}\right)(b - c)y = b - c$
> by assumption $a \mid a(b - c) \Rightarrow \left(\frac{m}{d}\right) \mid \left(\frac{a}{d}\right)(b - c)$
> $\Rightarrow \left(\frac{m}{d}\right) \mid \left[\left(\frac{a}{d}\right)(b - c)x + \left(\frac{m}{d}\right)(b - c)y\right] = b - c$

2.6.3. Inverse

The inverse of a modulo m ($a^{-1} \operatorname{mod} m$) is the smallest positive integer s.t. $ac \equiv 1 (\operatorname{mod} m)$

- $a^{-1} \operatorname{mod} m$ exists iff $a \perp m$
- the solution to linear equation $ax \equiv b (\operatorname{mod} m)$ is $x \equiv a^{-1}(ax) \equiv a^b (\operatorname{mod} m)$

Algorithm (matrix version)

```
set [r0 x0; r1 x1] = [a 1; m 0]
while r1 ≠ 0
    set [r0 x0; r1 x1] = [0 1; 1 −⌊r0/r1⌋] [r0 x0; r1 x1]
if r0 = 1
    return x0 [= a^{-1} mod m]
else return "no inverse"
```

$$\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} a & 1 \\ m & 0 \end{bmatrix}$$

$$\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{r_0}{r_1} \rfloor \end{bmatrix} \begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix}$$

return $x_{0_{[=a^{-1} \operatorname{mod} m]}}$

Example - solve $160x \equiv 7 (\operatorname{mod} 841)$

$$\begin{bmatrix} r_0 & a_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} a & 1 \\ m & 0 \end{bmatrix} = \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{160}{841} \rfloor \end{bmatrix} \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix} = \begin{bmatrix} 841 & 0 \\ 160 & 1 \end{bmatrix}$$

$\ldots$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{4}{1} \rfloor \end{bmatrix} \begin{bmatrix} 4 & -21 \\ 1 & 205 \end{bmatrix} = \begin{bmatrix} 1 & 205 \\ 0 & -841 \end{bmatrix}$$

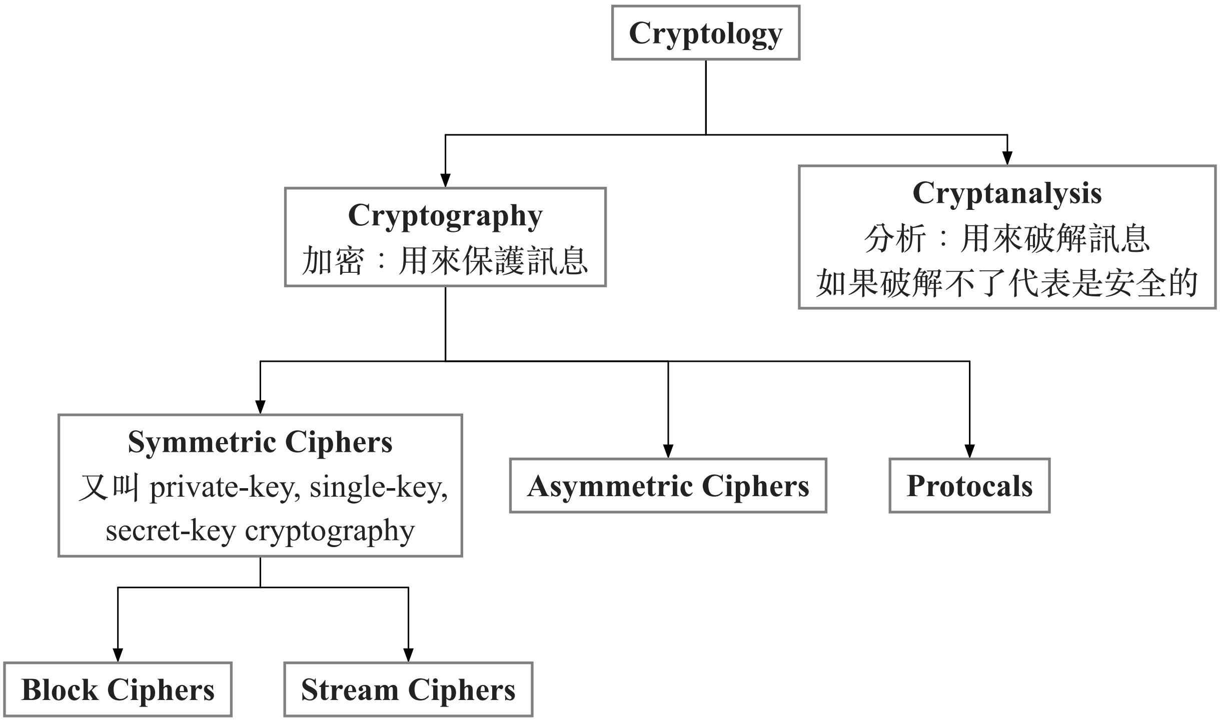
$\Rightarrow 160^{-1} \equiv 205 (\operatorname{mod} 841)$

$\Rightarrow x \equiv 160^{-1} \times 7 \equiv 205 \times 7 \equiv 594 (\operatorname{mod} 841)$

3. Finite Fields

4. Cryptography

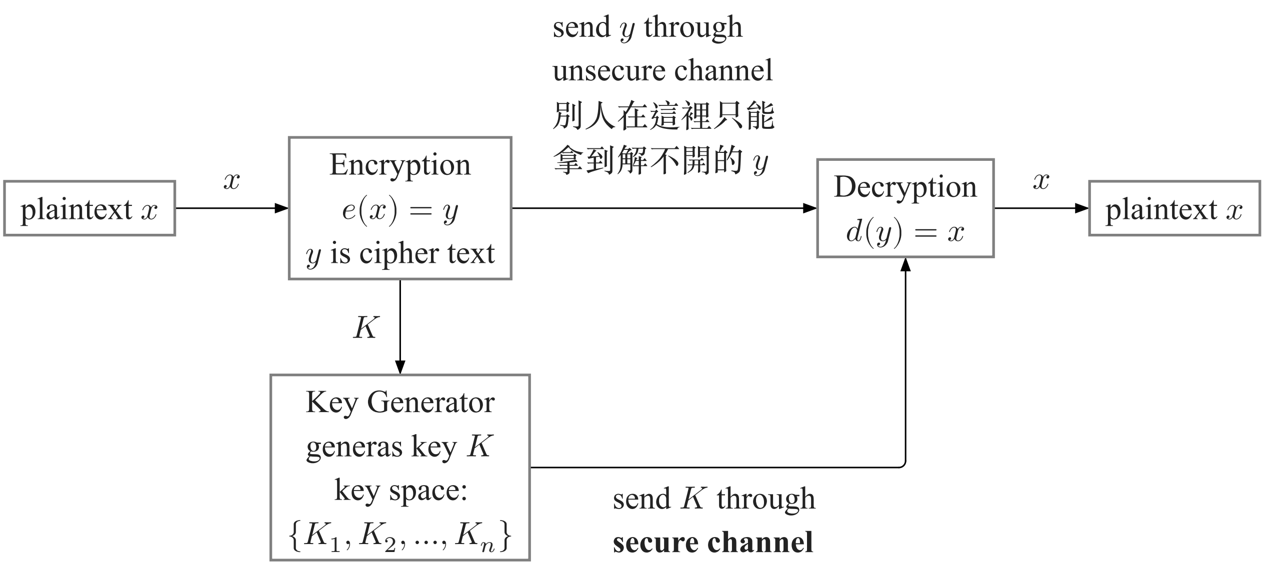
4.1. Introduction



Basic Fats

- Ancient Crypto: 2000 B.C. 古埃及 Letter-based encryption scheme 就很紅 (Ceasar cipher)
- Symmetric Ciphers: 從古時候到 1976 的加密都是 Symmetric 的
- Asymmetric Ciphers: 1976 Diffie, Hellman, Merkle 發明了 public-key cryptography
- Hybrid Schemes: 現在用的 protocal 大部分是都有用的
  - symmetric ciphers for encryption and message authenticatiohn
  - asymmetric ciphers for key exchange and digital signature

4.1.1. Symmetric Cryptography



encryption equation and decryption equation are inverse operations: $d_K(y) = d_K(e_K(x)) = x$

4.1.2. Cryptanalysis

Cryptanalysis

- **Classical Cryptanalysis**
  - **Brute-Force Attacks**
  - **Mathematical Analysis**
- **Implementation Attacks**
  Try to extract key through reverese engineering or power measurement
- **Social Engineering**
  例如用騙的

Brute-Force Attacks

key length	key space	Security life time
64	2^{64}	short term (幾天)
128	2^{128}	long-term (沒量子電腦的話幾十年)
256	2^{256}	long-term (有量子電腦也很安全)

- 現在還沒有量子電腦
- 如果有其他很簡單的攻擊方法的話很大的 key space 也沒用

4.1.3. Subsitution Cipher

方法：把每個字母都選一個特定的字母換掉
有兩種解法：

Exhaustive Key Search (Brute-Force)	**Letter Frequency Analysis** (Brute-Force)
一個一個找 有 $26 \times 25 \times \ldots \times 3 \times 2 \times 1 = 2^{88}$ keys subsitution table (=keys) 超大	出現頻率最高的英文字母是 e (大概 13%) 方法： 1. 用 e 換掉在 cipher text 裡面出現頻率最高的字母 2. 再用頻率次高的字母換掉 cipher text 裡面出現第二多次的字母 $\Rightarrow$ 要試的次數少很多

4.1.4. Shift (Ceasar) Cipher and Affine Cipher

兩種方法都會用把字母對應到數字 ⇒ A = 0、B = 1、…、Z = 25

Shift (Ceasar) Cipher

把每個字母用它 + key mod 26 換掉

Let $k, x, y \in \{0, 1, ..., 25\}$

- Encryption: $y = e_k(x) \equiv x + k \bmod 26$
- Decryption: $x = d_k(x) \equiv y - k \bmod 26$

key space is only 26

Affine Cipher

比 Shift Cipher 更高級一點

key : (a, b)

Let $a, b, x, y \in \{0, 1, ..., 25\}$

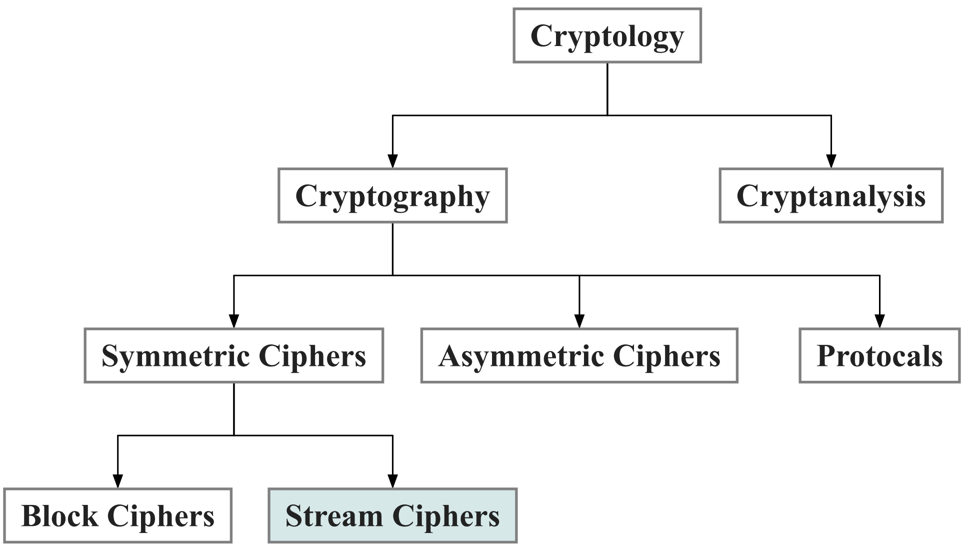
- Encryption: $y = e_k(x) \equiv ax + b \bmod 26$
- Decryption: $x = d_k(x) \equiv a^{-1}(y - b) \bmod 26$

a 只能選 $\gcd(a, 26) = 1$ 的 不然沒有 inverse (總共有 12 個)

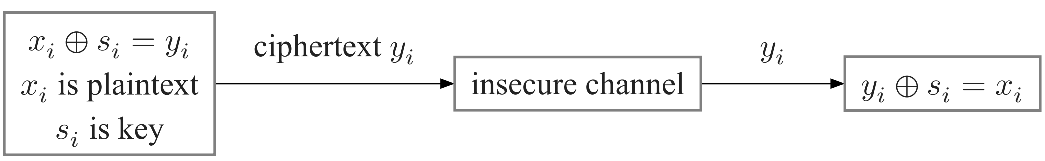
key space is $12 \times 26 = 312$

4.2. Stream Ciphers

4.2.1. Introduction to Stream Ciphers



Stream Ciphers	Block Ciphers
一次只加密一個字元 small and fast ⇒ common in embedded devices (e.g. A5/1 for GSM phones)	一次加密一整個 block are common for Internet applications



特性：

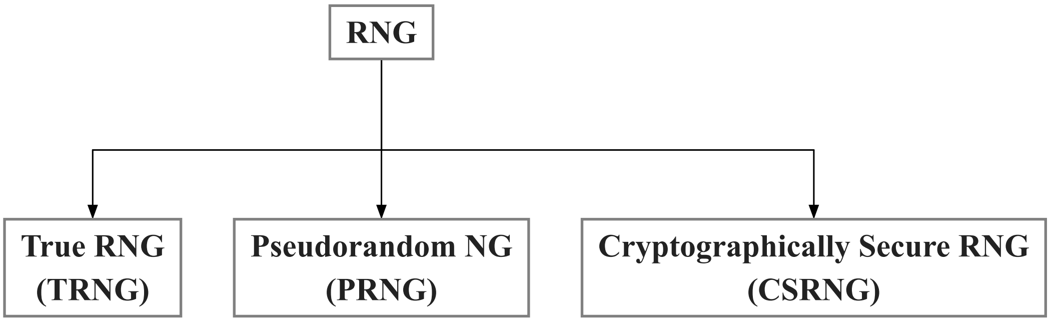
- XOR 的運算 (⊕) 跟加起來 mod 2 是一樣的 ⇒ 很簡單
- encryption and decryption are the same functions / inverting XOR is same as XOR itself $(x_i, y_i, s_i \in \{0,1\})$:
  - encryption : $y_i = e_{s_i}(x_i) = x_i + s_i \bmod 2$
  - decryption : $x_i = d_{s_i}(y_i) = y_i + s_i \bmod 2$
- output bit 是 0 或 1 的機率都是 50% ⇒ good statistic property

Synchoronous Stream Cipher	**Asynchoronous Stream Cipher**
Key stream depends only on the key (and possibly initialization vector)	Key stream depends also on the ciphertext 也就是一個位元形成密文之後會丟去 key generator 來生成後面的 key 所以後面的密文跟前面的密文是有關的

不同 symmetric cipher 的 performance 比較：

Cipher	**Key Length**	**Mbit/s**
DES	56	36.95
3DES	112	13.32
AES	128	51.19
RC4 (stream cipher)	(choosable)	211.34

4.2.2. Random Number Generators (RNGs)



TRNGs

- based on physical random processes (擲骰子或硬幣這種)
- output stream s_i should have good statistical properties ($Pr(s_i = 0) = Pr(s_i = 1) = \frac{1}{2}$)
- output 不能被 predict 或 reproduce
- used to generate keys or nonces (used only-once values), etc.

PRNG

- generate sequence form initial seed value
- output stream has good statistic properties
- output can be reproduced and can be predicted (**bad cryptographic properties**)
- computed in recursive way :
  - $s_0 = \text{seed}$
  - $s_{i+1} = f(s_i, s_{i-1}, ..., s_{i-t})$

> e.g. Simple PRNG : **Linear Congruential Generator**
> $S_0 = \text{seed}$
> $S_{i+1} = AS_i + B \bmod m$
> $\Rightarrow$ Assume A, B, S_0 are unknown and all 100 bits
> 300 bits of output are known (S_1, S_2, S_3)
> $\Rightarrow$ 只要解聯立方程式就有 A, B 了
> $S_2 = AS_1 + B(\bmod m)$
> $S_3 = AS_2 + B(\bmod m))$

CSPRNG

- special PRNG with additional property
  - **output must be unpredictable**
  - given n consecutive bits of output s_i, the following output bits S_{n+1} can not be predicted (in polynomial time)
- need in cryptography, in particular for stream ciphers
- 除了 cryptography 幾乎沒有其他 application 需要 unpredictability (但很多需要 PRNG)

4.2.3. One-Time Pad (OTP)

Unconditionally Secure Cryptosystem : A cryptosystem is unconditionally secure if it can't be broken with *infinite* computational resources

One-Time Pad

Let the plaintext, ciphertext, and key consist of individual bits $x_i, y_i, k_i \in \{0, 1\}$

- encryption : $e_{k_i}(x_i) = x_i \oplus k_i$
- decryption : $d_{k_i}(y_i) = y_i \oplus k_i$

OTP is unconditionally secure iff the key k_i is used once :

$$y_0 = x_0 \oplus k_0$$
$$y_1 = x_1 \oplus k - 1$$
$$\vdots$$

Every equation is a linear equation with two unknowns
$\Rightarrow$ for every $y_i, x_i = 0$ and $x_i = 1$ are equiprobable (等機率)
$\Rightarrow$ this is true iff $k_0, k_1, ...$ independent (k_i 的生成是完全隨機的)
$\Rightarrow$ it can be shown that this systems can *provably* not be solved
缺點 : 很不切實際因為 key 跟 message 要一樣長

4.2.4. Linear Feedback Shift Registers (LFSRs)

4.2.5. Trivium - A Modern Stream Cipher